

Food Borne Diseases Part 2

Viral foodborne disease: polio, hepatitis A, hepatitis E

NUT401 taken source: uploaded lecture

Poliomyelitis: agent and spread

- Acute infectious viral disease caused by poliovirus; may cause severe complications including paralysis.
- Three serologic types: type I epidemic/paralytic, type II endemic, type III less frequent.
- Humans are the reservoir; asymptomatic and symptomatic carriers spread infection.
- Contaminated food and water act as temporary secondary reservoirs in poor sanitation.

Polio pathogenesis

- Entry by fecal-oral route, replication in oropharynx and gastrointestinal tract.
- Invades local lymphoid tissue, may enter blood, then infects CNS cells.
- Motor neuron destruction produces characteristic flaccid paralysis.
- Incubation period: 7 to 14 days.

Outcome Of Polio Infection

Outcome	Frequency / key point
Asymptomatic	About 70% of child infections.
Abortive polio	About 24%; minor nonspecific illness.
Non-paralytic	Aseptic meningitis in 1% to 5%; recovery common.
Paralytic	<1%; spinal 79%, bulbar 19%, bulbospinal 2%.
Fatality / late	2% to 10% from respiratory paralysis; post-polio syndrome after 15-20 years.

Polio clinical picture and diagnosis

- Abortive disease: mild flu-like symptoms, URI, diarrhea, fever, sore throat, malaise.
- Non-paralytic: flu-like illness plus neck/back stiffness; may have nausea, vomiting, abdominal pain.
- Paralytic manifestations begin 7-10 days after onset; asymmetric flaccid paralysis, legs more than arms.
- Diagnosis: clinical picture plus virus from stool, CSF, or oropharyngeal secretions; serum antibodies.

Sabin OPV vs Salk IPV

Feature	OPV	IPV
Type	Live attenuated	Inactivated
Route	Oral	Injection
Immunity	Local + systemic	Systemic
Herd effect	Promotes herd immunity by excretion	Less effect on transmission
Main concern	VAPP / vaccine-derived outbreaks	No VAPP risk

Prevention, control, eradication

- Prevention: environmental sanitation, health education, vaccination.
- Egypt routine immunization: seven doses from birth to one and a half years.
- Control: notification, isolation, symptomatic treatment, contacts under 14-day surveillance.
- Eradication needs high OPV coverage, AFP surveillance, mass campaigns, mopping-up, cold-chain assurance.

Hepatitis A: essentials

- Highly contagious liver infection caused by HAV, a non-enveloped RNA virus of Picornaviridae.
- Transmitted fecal-orally by contaminated food/water or direct contact.
- Developing countries: early-life infection often few/no symptoms; developed countries: acute hepatitis in adolescents/adults.
- No chronic infection; usually full recovery with lifelong immunity; rare fulminant hepatitis.

HAV disease course

- Incubation: about 2-6 weeks, average 28 days.
- Prodrome: fatigue, nausea/vomiting, RUQ abdominal pain, fever, appetite loss.
- Icteric phase: jaundice, dark urine, pale stool, itching.
- Diagnosis: anti-HAV IgM for acute/recent infection; anti-HAV IgG for past infection/vaccination; ALT/AST and bilirubin may rise.

HAV prevention and HEV

- HAV prevention: safe water, sewage disposal, hand hygiene, safe food, food-handler hygiene.
- HAV vaccine: inactivated vaccines such as Havrix/Vaqta; combined HAV/HBV vaccine exists; 2-dose schedule with booster 6-18 months later.
- Passive immunoprophylaxis: immune globulin within 2 weeks after exposure, 0.02 mL/kg.
- HEV: fecal-oral, commonly waterborne; can be zoonotic; incubation 2-9 weeks; severe in pregnancy and immunocompromise; vaccine mainly available in China.